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Thermal conductivity of carbon nanotube and hexagonal boron nitride polymer composites

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ABSTRACT

This paper investigates the thermal conductivity of nanocomposites of carbon nanotubes (CNT) and hexagonal boron nitride (hBN) polymer composites using a three-dimensional random walk algorithm and effective medium approach (EMA) method. For CNT nanocomposites, the influence of the orientation of CNTs (semi-aligned and randomly-dispersed) on the thermal conductivity was quantified. For hBN nanocomposites, thermal conductivity values were studied in random dispersion of inclusions. Hybrid nanocomposites containing both hBN and CNT were also investigated using a random walk method and experiments. To verify the calculated thermal conductivities, nanocomposite samples were fabricated. Both the modeling and experimental results showed that the concentration of nanoparticulates (random dispersion) in composites increases thermal conductivity up to 170% and 300% for 13.1 vol% CNT and 6.3 vol% hBN, respectively. It was found that the alignment of CNTs may enhance thermal conductivity of CNT composites. For composites with a 13.1 vol% of CNTs, the increase in thermal conductivity of the semi-aligned composites is approximately 210%. However, for hybrid composites with a 3.1 vol% of CNT/hBN (1.55 vol% each), a 290% increase is observed. At 3.1 vol%, thermal conductivity enhancements are 170% and 250% for single-filled composites of CNT and hBN, respectively.

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1. Introduction

In recent years, thermally and electrically conductive nanocomposites have been widely studied in academia and industry due to their outstanding multifunctional properties. High performance nanocomposites have been fabricated utilizing one-dimensional nanoparticulate fillers which include carbon nanotubes (CNT), carbon nanofibers and metal nanowires [1–4]. CNTs have unique physical, electrical, and thermal properties, which render nanocomposites attractive in various fields [5]. The thermal conductivity of a single carbon nanotube was expected to reach up to 3000 W/mK at room temperature based on molecular dynamics (MD) models, which is much higher than regular steel with approximately 50 W/mK [6]. This high thermal conductivity of CNTs is anticipated to enhance the thermal properties of polymers (with thermal conductivity of less than 1 W/mK for common polymers) when they are used as a filler. However, the observed thermal conductivity of polymer composites is not as high as expected. High concentrations of CNT with more than 30 vol % in nanocomposites produce relatively appropriate thermal conductivity ranging around 20 W/mK. However, high loadings of fillers lead to significant processing challenges [7].

One of the common methods to enhance thermal conductivity of CNT nanocomposites is to fabricate nanocomposites with aligned CNTs. Aligned CNT polymer nanocomposites are applicable in various areas such as thermal managements in electronics [8]. Mahmoodi et al. fabricated nanocomposites with relatively well-controlled morphology through semi-aligned CNTs using injection molding [9]. They reported that having semi-aligned CNTs within a polymer results in higher thermal conductivity values in the axial direction and lower values in the transverse direction. This could be explained by the fact that thermal transport within the materials is dominated by phonons (lattice vibration); at boundaries (between CNTs and matrix), high-frequency phonon modes in a CNT need to be transferred to low-frequency modes through phonon-phonon coupling in order to be transported to the

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surrounding medium. This acoustic mismatch occurs more in the transverse direction and results in lower thermal conductivity [10].

The term "semi-aligned" is used in this paper to describe nanocomposites with lower levels of CNT orientations within a matrix compared to perfectly aligned ones. The term "aligned" is used by various researchers to describe CNT nanocomposites with levels of alignment much higher than that of the ones presented in this paper [8,11,12]. Similar to this paper, the term "semi-aligned" for CNT composites is used to describe lower levels of alignment compared to perfectly aligned CNT compositions [13–15]. The procedure to fabricate composites with oriented CNTs is an established technique, which has been conducted and reported in previously published research [9,16–19]. In this method, a micro-injection molding technique is used to create high shear rates during injecting to orient the CNTs.

Hexagonal boron nitride (hBN), another highly thermally conductive nanoparticulate similar to that of graphene layers, is the most stable and softest among the BN polymorphs. Like graphite, boron and nitrogen atoms are bound together by strong covalent bonds within each hBN layer, forming an hBN sheet, and a weak van der Waals force occurs between the layers [20,21]. Films made of this material have high-temperature thermal conductivity and stability with high mechanical strength and high corrosion resistance [22–25]. Because hBN has good electrical insulation properties, it has also been applied as a charge barrier layer for electronic equipment [21]. hBN powders are used in various applications such as coatings, lubricants, ceramic parts, corrosion resistance in the steel/iron industry, and electrical insulation in the semiconductor industry. The addition of hBN to polymers increases thermal conductivity, reduces thermal expansion, and adjusts dielectric loss behavior of the composite [24].

Some researchers have tried to fabricate hybrid nanocomposites containing both CNTs and graphene nanoplatelets [26–28]. The challenge in developing these nanocomposites is to homogeneously disperse nanoparticulates with low agglomerations, due to their electrostatic interactions and van der Waals forces. Due to the synergistic effects of using CNT and graphene nanoplatelets together, the main purpose in combining these two nanoparticulates and fabricating hybrid nanocomposites is to enhance thermal and electrical conductivity. The synergism originates from bridging graphene nanoplatelets with CNTs, extending the conductive network of nanoparticulates. This leads to reducing the thermal interface resistance along the 1D-2D hybrid network [26]. Since hBNs are very similar to graphene nanoplatelets, it is expected that the behavior would also be similar to that of hybrid compositions of hBN and CNTs [29,30].

Numerical and theoretical methods are conducted to predict the thermal and electrical conductivities of the composites with highly thermally conductive particulates [31–38]. Classical approaches that estimate the thermal conductivity of composites, such as the Effective Medium Theory (EMT), do not take into account the contact resistance between the CNTs or the Kapitza resistance at the CNT/polymer interface [39]. Nielsen proposed a modified effective medium approach (EMA) to predict the thermal conductivity of two-phase systems with randomly-filled fibers [40]. However, Guthy and colleagues applied Nielsen's model to SWCNT/PMMA nanocomposites and reported that the model did not fit the experimental results well [41]. Nan et al. introduced a general equation to estimate thermal conductivity of two-phase composites, which can be used in a wide variety of particle geometries considering the thermal boundary resistance [42,43]. The EMA method with Kapitza resistance at CNT-polymer boundaries to predict the thermal conductivity of aligned CNTs has also been developed by Yamamoto et al. [10]. Duong et al. developed a new algorithm for calculating the thermal conductivity by taking into account the thermal boundary resistance at the interfaces between the matrix and CNTs [44]. However, they left CNT - CNT contact out of consideration, which is very common in the CNT nanocomposites [45].

The main objective of this paper is to investigate the effects of adding CNTs and hBN to a polymeric matrix (Polystyrene) on thermal conductivity. It is observed that the hybrid compositions of CNT/hBN exhibit greater thermal conductivity values compared to single-filled composites. This synergistic effect of CNTs and hBN on the thermal conductivity of composites is also studied in this paper. Investigating the thermal behavior of hybrid composites, taking into consideration the concentrations of the fillers and their geometries, could lead to optimized polymer composites with enhanced thermal conductivities [30]. A new 3D model based on a random walk method is developed to study the thermal conductivity of single-filled and hybrid composites. The thermal conductivity of nanocomposites is evaluated through modeling and experimentation by considering nanocomposites with different nanoparticulates (CNT and hBN), loadings, and orientations (random and semi-aligned). A theoretical model following the EMA method is also employed to compare the experimental and random walk model results with theoretical results.

Many researchers have attempted to model complex CNT networks, and, to date, mostly 2D models have been reported [46,47]. There are a variety of qualitative 3D models for the physical properties of nanocomposites, but quantitative models for large CNT networks have not yet been thoroughly studied [48]. In addition, a comprehensive 3D representative volume element (RVE) to capture the synergistic effects of the CNTs and hBNs' characteristics, which are sensitive enough to differ among composites with small changes in microstructure, is absent [49]. In this paper, the developed 3D model can fulfill this gap in the research.

The model presented in this paper is an extension of the authors' comprehensive modeling of nanocomposites' behavior based on a random walk method. The first part of the research is on predicting electrical properties considering the effects of foaming, CNT breakage and alignments, and force sensing capabilities of CNT nanocomposites [4,36]. The second part, which is presented in this paper, is focused on developing a new model to predict the thermal conductivity of CNT/hBN composites considering the different characteristics of the fillers.

Combining these two parts (electrical and thermal conductivity predictions), one can easily predict both the electrical and thermal properties of a nanocomposite structure. In addition, using a single model for both electrical and thermal properties results in sharing one RVE, which leads to a reduced runtime. Hence, nanocomposites with a large number of nanoparticulates, similar to real conditions, could be investigated using the developed comprehensive model. The developed model presented in this paper has the above mentioned advantages compared to models developed in previously conducted research [48].

The remainder of this paper is organized as follows: first, the developed random walk computational model for thermal conductivity is described, and the relevant simulation parameters are presented; second, the used effective medium approach is introduced with definitions of the required parameters; third, the equipment, materials, tests and standards used for the thermal conductivity measurements are presented in the experimental section; finally, the results obtained from these three different methods are compared and discussed.

2. Modeling methods

Several models have been reported to predict thermal conductivity of composites with different particulates including

continuous fibers, laminated flat plates, spheres and ellipsoidal particles [7]. For CNT and polymer composites, concentrations [50], interfacial resistance [51], and aspect ratio [52] of fillers have been studied. The present paper aims to develop a stochastic model to predict thermal conductivities using three-dimensional random walk algorithm and the effective medium approach (EMA) method for various nanoparticulates and contents including hybrid compositions.

2.1. Random walk algorithm for thermal conductivity

The thermal conductivity of the composite can be computed from an off-lattice Monte Carlo simulation [53,54], in which a large number of random thermal walkers were traveling in the computational representative volume element (RVE) until a steady state is achieved. In the present study, the numerical simulation is carried out on a 10 × 10 × 10 μm RVE with polystyrene (PS) and CNT phases. The same RVE is used for other two types of nanocomposites, hBN/PS and hBN/CNT/PS, which are shown schematically in Fig. 1. The RVE is divided into bins and a temperature profile for the RVE is constructed based on the number of walkers in each bin.

CNTs and hBNs, whose shapes are one- and two-dimensional respectively, are dispersed in the matrix. For CNTs, two different dispersion conditions are considered: randomly-oriented and semi-aligned CNTs. Heat flux is applied constantly in one direction, parallel to x-axis. Hot walkers are injected from one side (e.g., the plane at $x = 0$ μm) of the RVE, and an equal number of cold walkers (carrying negative energy) are injected simultaneously from the other side (e.g., the plane at $x = 10$ μm). Hot walkers can exit only at the plane located at x = *cube length* (i.e., x = 10 μm) and are reflected off from the other planes. Similarly, cold walkers can exit only at $x = 0$ plane and are reflected off from the other planes. Since no heat can be accumulated in the RVE, it reaches a steady-state by interactions between hot and cold walkers. Each simulation is carried out for 100 ns, where 90,000 walkers, both hot and cold walkers, are released and distributed uniformly on the respective plane surfaces at every time step.

The travel of walkers in the matrix can be described by the Brownian motion. The motion is defined by a normal distribution with a mean of zero and a standard deviation that depends on the matrix thermal diffusivity, D_m. The standard deviation (σ) of the distribution in each space is given by:

$$\sigma = \sqrt{2D_m\Delta t} \tag{1}$$

where Δt is the time increment. Since the nanoparticulates (CNTs and hBNs) used in this paper are assumed to have very high thermal conductivity values, the thermal walkers move in jumps that follow a uniform distribution within the nanoparticulates. It is noted that the thermal resistance for a walker traveling from the matrix to a nanoparticulate equals that from a nanoparticulate to the matrix, as required by the principle of microscopic reversibility. With this assumption, once a walker in the matrix reaches the interface between the matrix and a nanoparticulate, the walker will move into the nanoparticulate phase with a probability of $f_{m\text{-}filler}$ ($f_{m\text{-}CNT}$ or $f_{m\text{-}hBN}$) which represents the thermal resistance of the interface (and will stay at the previous position in the matrix with a probability $1 - f_{m-filler}$). Accordingly, the program generates a probability at random and compares this with $f_{m\text{-}filler}$. If this number is greater than $f_{m\text{-}filler}$, then the marker remains at the interface; otherwise, it moves into the inclusion (CNT or hBN).

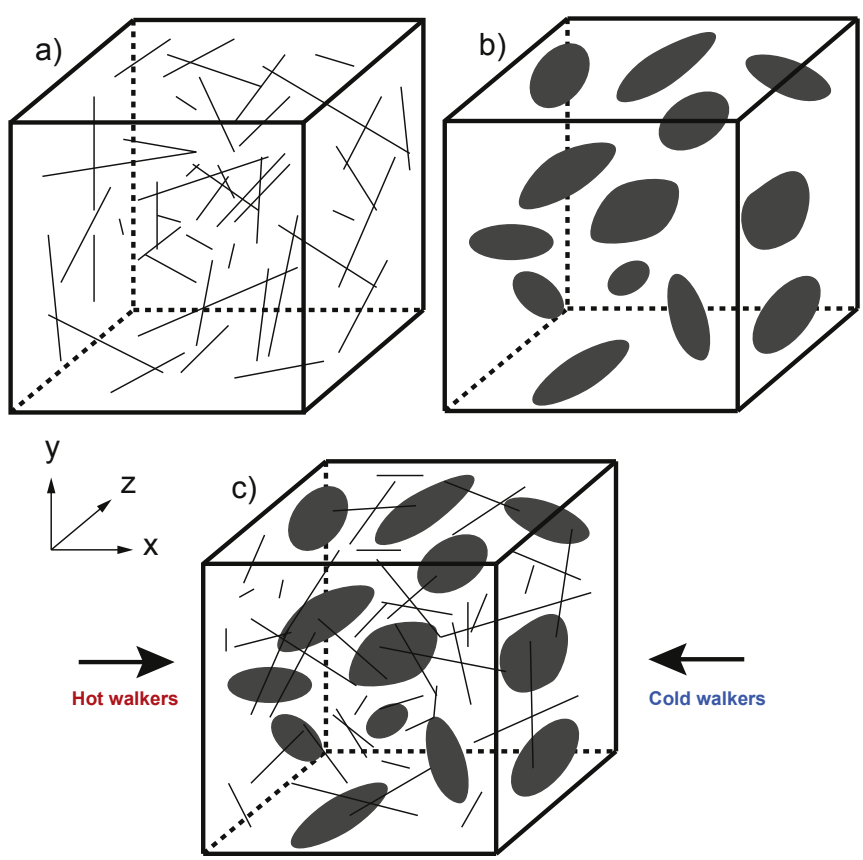

Fig. 1. Schematics of (a) randomly oriented CNTs, (b) hBN, and (c) hybrid within RVEs.

Table 1
The simulation parameters for modeling.

Parameter	Value
Number of grids per direction	96
Total number of walkers	90,000
Total time	100 ns
Time increment	0.0025 ns

Similarly, once a walker is inside an inclusion, the walker will redistribute randomly within the inclusion with a certain probability ($1\text{-}f_{filler\text{-}m}$) at the end of a time step and will cross into the matrix phase with a probability of $f_{filler\text{-}m}$. Accordingly, program generates another probability value at random and compares this with $f_{filler\text{-}m}$. If this number is greater than $f_{filler\text{-}m}$, then the marker moves within the inclusion, or else it moves into the matrix.

While moving inside the inclusion, the markers move towards one of the end points of the inclusion in half the time and in the opposite direction for the remaining time. In order to move into the matrix, the marker first moves to any point on the inclusion surface and then proceeds in the matrix by Brownian. Thermal equilibrium is maintained by:

$$f_{filler-m} = \frac{C_0 \sigma A_c}{V_c} f_{m-filler} \tag{2}$$

where A_C and V_C are the surface area and volume of the CNT or hBN, respectively.

According to the acoustic mismatch theory, the average probability for transmission of phonons across the interface into the dispersion, $f_{m\text{-}filler}$, is given by Ref. [55]:

$$f_{m-filler} = \frac{4}{\rho C C_m R_{bd}} \tag{3}$$

where ρ is the matrix density, C is the matrix specific heat, C_m is the velocity of sound in the matrix, and R_{bd} is the thermal boundary resistance at the CNT/polymer or hBN/polymer interface.

2.1.1. Parameters of the composites used in the simulation

The heat flow is studied in a 10 μm size cubic RVE containing nanotubes with a 10 nm diameter and ellipsoidal hBNs with a 100 nm thickness. The CNTs and hBNs are dispersed within the RVE and have an average length of 1000 nm. The orientation of CNTs is divided into two categories: semi-aligned and random dispersion. hBN nanoparticulates are randomly dispersed. The thermal conductivity of the polymeric matrix was assumed to be 0.14 W/mK and that of CNT and hBN to be 200 and 300 W/mK, respectively [9,56,57]. The modeling parameters are also tabulated in Tables 1 and 2.

Zhong et al. reported systematic MD studies of the effect of contact morphology on the CNT - CNT contact resistance [58]. The reported values of the CNT - CNT contact resistance are in order of 10^{-8} m^2 K/W [7,51]. As the CNTs can be functionalized to reduce their contact resistance, CNT line contact is assumed and the CNT - heat source thermal boundary resistance is assumed to be equal to the CNT - matrix thermal boundary resistance for simplicity.

The simulations are conducted with two different orientations of CNTs dispersed in composites (semi-aligned and random). Prior to this stage, using different thermal boundary resistance and contact resistance values, the simulation process is optimized considering different CNT concentrations (1, 4 and 8 vol%). The optimization is done through comparing previously published research works [7,42,59–64].

Fig. 2 illustrates an example of a simulation run for the nanocomposite with 0.1 vol% fraction of CNTs and hBNs, dispersed in a 10 × 10 × 10 μm cube. The computational cell is heated from one surface (the $x = 0$ plane) with the release of 90,000 walkers (cold and hot walkers) distributed randomly on the surface. The temperature distribution is calculated from the number of walkers found in each bin. The walkers move through the matrix material by Brownian motion. The Brownian motion in each space direction is simulated with random jumps that each walker executes at each time step (time increment of 0.0025 ns). CNT - CNT thermal boundary resistance (R_{bd}), and CNT – polymer boundary resistance were taken to be 10^{-8} m^2 K/W and 10×10^{-8} m^2 K/W, respectively. For other contact resistances, hBN – hBN, and hBN polymer, 7×10^{-10} m^2 K/W, and 10×10^{-8} m^2 K/W were used. Modeling steps are illustrated in Fig. 2 (a), where yellow boxes indicate random stages. Fig. 2 (b) and (c) show two stages of the modeling, random distribution of nanoparticulates and final distribution of walkers. In Fig. 2 (c), hot walkers are depicted in red (left side) and the cold walkers in blue (right side), and the grid lines are shown in green.

2.2. Effective medium approach (EMA)

EMA method is widely used for thermal conductivity modeling of nanocomposites containing CNTs and hBN. There are several models for predicting thermal conductivities of two-phase compositions. Several researchers have proposed a variety of analytical models, which are based on Fourier's law of heat transfer [65]. For CNT/polymer composites, the value of 20 W/mK is used in theoretical modeling of the thermal conductivity of CNTs. This value is used for reducing the mismatch between EMA values and experimental results. A similar K_{CNT} value is used by some researchers [9,10]. Contact resistances for CNT - matrix and CNT - CNT are assumed to be 10^{-7} m^2 K/W and 10^{-8} m^2 K/W, respectively and for hBN - hBN and hBN - matrix, 7×10^{-10} m^2 K/W and 10×10^{-8} m^2 K/W are used, respectively [10].

2.2.1. CNT/polymer thermal conductivity EMA modeling

In the present paper, an EMA model is implemented to include the thermal contact resistance and partial alignment of CNTs in thermal conductivity modeling of nanocomposites. Complete description of the EMA model is presented elsewhere [39,42,43]. Due to the large aspect ratio of CNTs in this study, the axial $\left(K_{\amalg}\right)$ thermal conductivity of the nanocomposites is reduced to the following nonlinear expressions [9,10,39,43]:

$$\left(K_{\amalg}\right) = A_{\amalg} + \frac{C_{\amalg}}{B_{\amalg} - f} \tag{4}$$

where

$$A_{\amalg} = \frac{K_m\left(a_1(-1+b)(1+b\cos^2\theta) + b(-1+b\cos^2\theta)K_m R_{bd}\right)}{(-1+\cos^2\theta)\left(a_1(-1+b) - bK_m R_{bd}\right)} \tag{5}$$

$$B_{\amalg} = -\frac{a_1 + a_1 b + bK_m R_{bd}}{(-1+\cos^2\theta)\left(a_1(-1+b) - bK_m R_{bd}\right)} \tag{6}$$

$$C_{II} = \frac{K_m(a_1 + a_1 b + bK_m R_{bd})(a_1(-1+b)(2+(-1+b)\cos^2\theta) + b(-2+\cos^2\theta + b\cos^2\theta)K_m R_{bd}}{(-1+\cos^2\theta)^2(a_1 - a_1 b + bK_m R_{bd})^2} \tag{7}$$

where f is the CNT volume fraction, K_m is the thermal conductivity of the polymer matrix [W/mK], a_1 is the radius of the nanotube [m], b is defined as K_{CNT}/K_m, and R_{bd} is the Kapitza resistance at the CNT-polymer boundaries, 10^{-7} [Km^2/W] [7,51,66]. The properties of the utilized CNTs are shown in Table 3.

$<cos^2\theta>$ is defined as the degree of CNT orientation in the following equation:

$$<\cos^2\theta> = \frac{\int \rho(\theta)\cos^2\theta \sin\theta d\theta}{\int \rho(\theta)\sin(\theta)d\theta} \tag{8}$$

where θ is the azimuthal angle, with statistical distribution among all CNTs, $\rho(\theta)$, as the weighting factor. For perfect alignment,

Table 2
RVE, CNTs, hBNs and matrix's properties used in simulation.

Property	Value	Property	Value
RVE volume	10 × 10 × 10 μm	Thermal equilibrium factor (C_0)	0.25
CNT diameter (average)	10 nm	Matrix thermal conductivity	0.14 W/mK
CNT aspect ratio	100	Matrix density	1050 kg/m^3
hBN thinness	70 nm	Matrix specific heat	1.3 kJ/kg K
hBN length	1000 nm	Matrix thermal diffusivity	10.5 m^2/s
hBN width	500 nm	Step size of marker in matrix (σ)	1.6176 nm

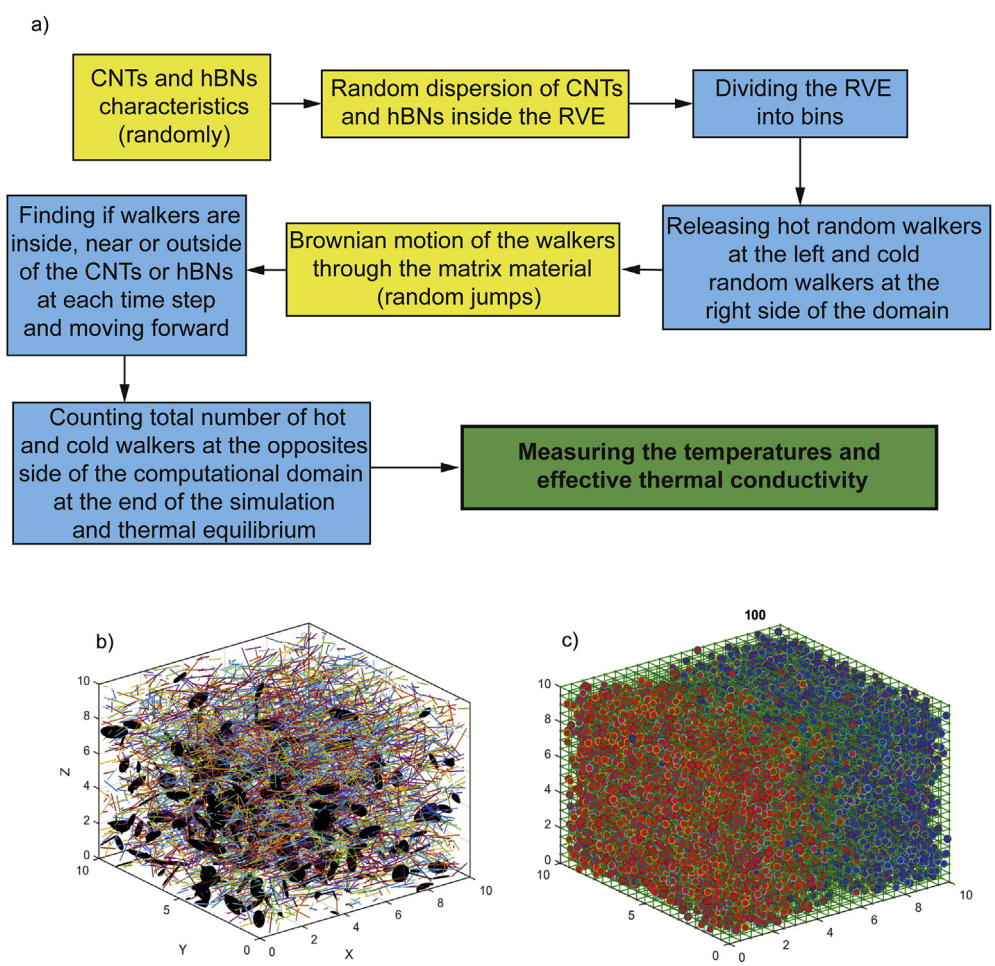

Fig. 2. (a) Modeling stages and schematic illustration of (b) randomly distributed CNTs and hBNs within the RVE and (c) final distribution of walkers in the RVE.

Table 3
Parameters used in EMA.

CNT length	1 μm
CNT radius (A_1)	10 nm
Degree of orientation (<cos^{2}θ>)	From Fig. 3
CNT volume fraction (f)	0–13.04
PS thermal conductivity (K_m)	0.14 (W/mK)

$<cos^2\theta> = 1$, and $<cos^2\theta> = 1/3$ for random distribution of CNTs. Considering the very small size and geometry of the CNTs, it is very difficult to characterize the orientations. For this reason, finding the azimuthal angle of the CNTs in a 3D structure with available experimental setups such as Raman spectroscopy or X-ray diffraction is extremely challenging. Therefore, transmission electron microscopy (TEM) images, which are in 2D, are used. A method described in previously published works [7,9] is used in performing this task. For each sample, 500 CNTs are considered in determining the mean square cosine from the TEM images.

2.2.2. hBN/polymer thermal conductivity EMA modeling

To theoretically calculate the thermal conductivity of the nanocomposites, the same model for CNT/polymer nanocomposites is used. For this purpose, it is assumed that the nanocomposites contain ellipsoidal inclusions with thermal interface resistance between nanoparticulates and the polymeric matrix. The thermal conductivity values of the nanocomposites with ellipsoidal nanoparticulates (hBN) can be calculated as follows [39,67]:

$$K_{\coprod} = K_m \frac{2 + f\left[\beta_{\coprod}\left(1 - L_{\coprod}\right)\left(1 + \langle Cos^2\theta\rangle\right) + \beta_{\perp}(1 - L_{\perp})\left(1 - \langle Cos^2\theta\rangle\right)\right]}{2 - f\left[\beta_{\coprod} L_{\coprod}\left(1 + \langle Cos^2\theta\rangle\right) + \beta_{\perp} L_{\perp}\left(1 - \langle Cos^2\theta\rangle\right)\right]} \tag{9}$$

$$K_{\perp} = K_m \frac{1 + f\left[\beta_{\coprod}\left(1 - L_{\coprod}\right)\left(1 - \langle Cos^2\theta\rangle\right) + \beta_{\perp}(1 - L_{\perp})\langle Cos^2\theta\rangle\right]}{1 - f\left[\beta_{\coprod} L_{\coprod}\left(1 - \langle Cos^2\theta\rangle\right) + \beta_{\perp} L_{\perp}\langle Cos^2\theta\rangle\right]} \tag{10}$$

where K_m and K are thermal conductivities of the matrix and composite, respectively. Subscripts for K denote the direction of conductivity measurement. L, f and θ are geometrical factor, volume fraction and angle between the composite axis and the inclusion symmetric axis, respectively. β is also defined in Equation (11):

$$\beta_{ii} = \frac{K_{ii}^{hBN} - K_m}{K_m + L_{ii}\left(K_{ii}^{hBN} - K_m\right)} \tag{11}$$

where K_{ii}^{hBN} is thermal conductivity of the inclusion with thermal interface resistance along axis X_i of the inclusion. For random oriented hBNs, the aspect ratio, $p = t/d$ is close to zero and $L_{II} = 0$ and $L_{\perp} = 1$ and $<Cos^2\theta> = 1/3$. So, thermal conductivity can be written as [67]:

$$\frac{K}{K_m} = \frac{3 + 2f\beta_{\coprod}}{3 - f\beta_{\perp}} \tag{12}$$

where

$$\begin{cases} \beta_{\coprod} = \dfrac{K_{\coprod}^{hBN}}{K_m} - 1 \\ \beta_{\perp} = 1 - \dfrac{K_m}{K_{\perp}^{hBN}} \end{cases} \quad \begin{cases} K_{\coprod}^{hBN} = \dfrac{K_{hBN,\coprod}}{1 + \dfrac{2R_{bdm}K_{hBN,\coprod}}{d}} \\ K_{\perp}^{hBN} = \dfrac{K_{hBN,\perp}}{1 + \dfrac{2R_{bdm}K_{hBN,\perp}}{t}} \end{cases} \tag{13}$$

Since $K_{hBN}/K_m \gg 1$, and f is small, Equation (12) can be simplified as:

$$\frac{K}{K_m} = 1 + \frac{2}{3}f\left[\frac{\left(\frac{K_{hBN}}{K_m}\right)}{1 + 2R_{bdm}\left(\frac{K_{hBN}}{d}\right)}\right] \tag{14}$$

where d, t and R_{bdm} are hBN's diameter and thickness and thermal interface resistance, respectively. Equation (14) is used to calculate the thermal conductivity of composites. To present and compare the results, ratio of thermal conductivity of composites to that of the matrix (PS) is reported.

3. Experiments

In this study, a masterbatch of 20 wt% PS/CNTs (Hyperion Catalysis) is used and diluted into nanocomposites with various CNT loadings (0.5, 1, 2, 3.5, 5 and 10 wt%) using a twin-screw extruder. Density values of CNTs and PS are used for preparing different concentrations. Multi-walled CNTs are vapor grown and have 10–15 nm in outer diameter, 1–10 μm in length and 1.75 g/cm^3 in density. hBN powder (MK-hBN-N70, M.K. Impex Corp.) with average size of 70 nm and purity of 99.5% is also used. Polystyrene (PS) (Styron® 610, American Styrenics LLC) with a density of 1.05 g/cm^3 and melt flow index of 11 g/10 min is used as the polymeric matrix. Thermal conductivity of the PS, represented by K_{ps}, is 0.14 W/mK, which is given by the manufacturer. All the materials for the experiments are dried for 4 h at 80 °C.

For hBN/PS nanocomposites, solution mixing is utilized. This technique uses chloroform as a solvent for PS with ratio of 10:1. The required amount of hBN (and CNT) is added to the solution. The solution is then sonicated for 20 min for dispersing the nanoparticulates into the matrix. A compression molding machine (CARVER, Model 4122) is used to fabricate rectangular samples (30 × 20 × 1.8 mm) for characterization. Specimens were compression-molded for 10 min, while the temperature of the platens and pressure were kept at 210 °C and 38 MPa, respectively. hBN/PS and hBN/CNT/PS are fabricated through the compression molding. CNT/PS specimens with randomly dispersed CNT are also fabricated though this technique. Table 5 summarizes the characteristics of the fabricated nanocomposites.

Table 4
Injection molding conditions.

	Mold temp. (°C)	Melt temp. (°C)	Holding pressure (bar)	Injection velocity (mm/s)
Semi-aligned	60	215	100	24
Regular	60	240	60	24

Table 5
Summary of the experiments.

CNT concentrations (vol%)	hBN concentrations (vol%)	Hybrid concentration, CNT + hBN (vol%)	Orientations	Mixing methods	Molding methods
0.3 (±0.01)	0.3 (±0.01)	0.3 + 0.3 (±0.01)	Semi-aligned	Melt mixing	Injection
0.61 (±0.01)	0.61 (±0.01)				
1.22 (±0.01)	1.22 (±0.01)				
2.15 (±0.01)	3.1 (±0.01)	1.55 + 1.55 (±0.01)	Random	Solution	Compression
3.1 (±0.01)					
6.3 (±0.01)	6.3 (±0.01)				
13.15 (±0.01)					

As summarized in Table 4, a micro-injection molding machine (BOY 12A) with a screw diameter of 18 mm and a length to diameter ratio of 20 is used for preparing the injection molded samples with processing conditions to fabricate nanocomposites with partially aligned CNTs. Optimal processing conditions for creating semi-aligned CNT composites is chosen based on a series of experiments conducted using a two-level, four-factor factorial design. Details of the optimizations can be found elsewhere [16]. Semi-aligned composites feature a relatively high degree of alignment than those of regular ones (random dispersed composites). This high degree of alignment is attributed to a low melting temperature, high viscosity, and holding pressure during the filling step in injection molding. A TEM system is employed to evaluate the alignment of the fabricated composites. A very thin film of

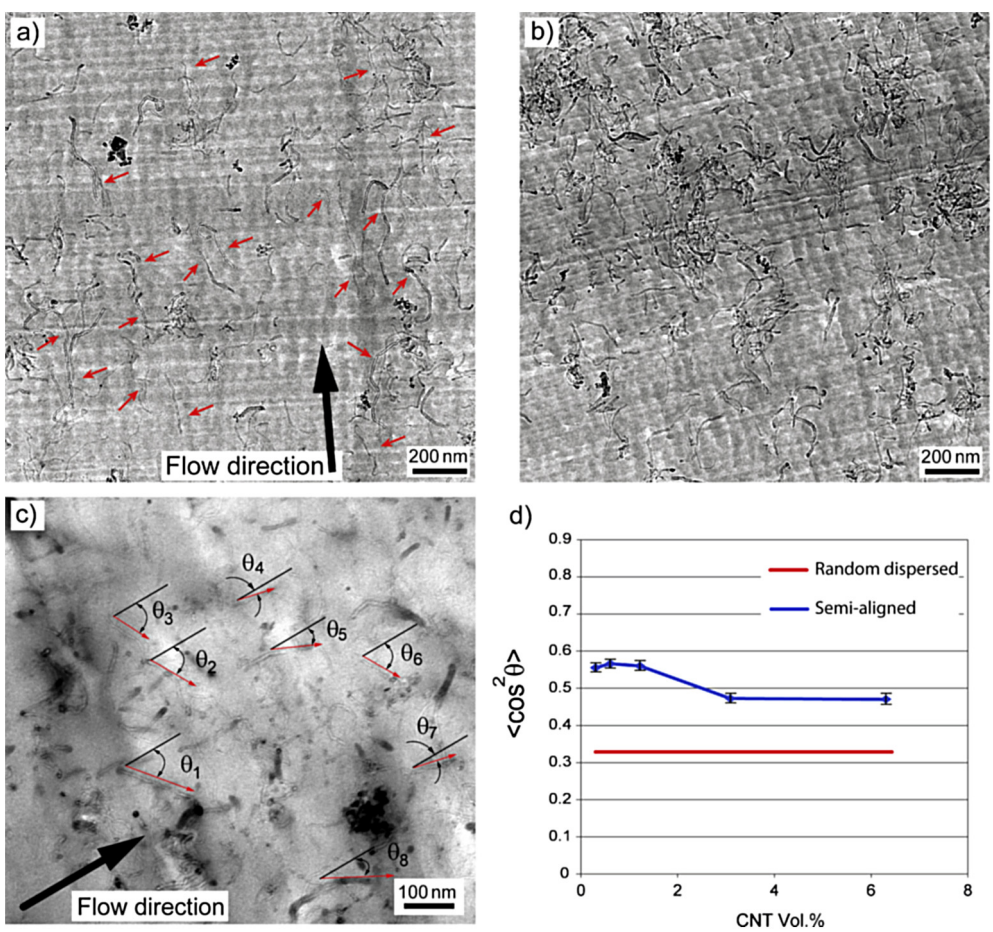

Fig. 3. Alignment characterization: TEM images of (a) semi-aligned injection molded composite (injection direction is indicated with large arrows and partially aligned CNTs with small arrows) and (b) randomly-dispersed CNTs in compression molded composite, (c) azimuthal angles of CNTs relative to the flow direction (small arrows represent CNTs direction, and straight lines and large arrow indicate direction of injection), and (d) mean-square cosine values of semi-aligned and random dispersed CNT composites as a function of the amount of CNT loadings.

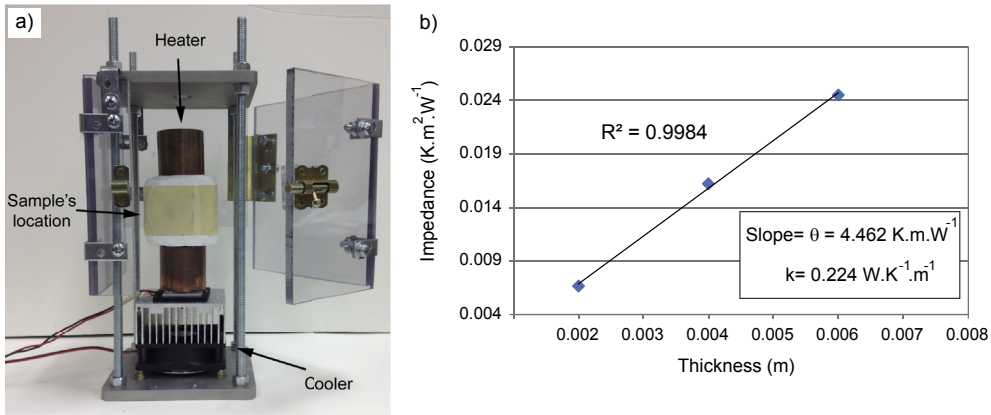

Fig. 4. (a) Thermal conductivity measurement device, (b) thermal impedance vs. thickness of specimens.

composites (<100 nm thickness) is cut, and the orientation of CNTs is investigated under the TEM [9,16,17,68–70].

Fig. 3 shows two different alignments in composites through TEM picturing of a very thin layer of specimens. Partially aligned CNTs are marked with small arrows in the TEM picture (Fig. 3 (a)), and no aligned CNTs are found in the randomly dispersed composites (Fig. 3 (b)). Since high shear stresses are applied to molten material during injection, the semi-aligned CNTs are parallel to the flow direction of injection which is indicated with a large arrow in the figure. On the other hand, Fig. 3 (b) exhibits the random dispersion of the CNTs within the polymer, as compression molding exerts very low unidirectional shear stress [9,16].

A quantitative method is used to evaluate the orientation of CNTs within the polymeric films. Since finding the azimuthal angle of CNTs in a 3D structure with Raman spectroscopy, SEM, and X-ray diffraction is very difficult, the azimuthal angles are measured from 2D TEM pictures (Fig. 3 (c)) [9]. In this method, for each composition, the orientation factor (mean-square cosine, $<Cos^2\theta>$) of 500 CNTs in different locations in the sample is calculated using Equation (8). As illustrated in Fig. 3 (c), some CNTs are not straight; in the case of a small curvature, the major direction of the CNTs is represented by a vector. The mean-square cosine values are calculated for the semi-aligned composites fabricated by injection molding, and the results are exhibited in Fig. 3 (d). The calculated mean square cosine values for the semi-aligned composites are higher than that of the randomly-dispersed CNT composites, ranging from 0.45 to 0.6. The mean-square cosine values of perfect alignment and random dispersion are known to be 1 and 0.33, respectively [7]. This method was previously used to prove the alignment of CNTs within the polymeric matrices [8,9].

The thermal conductivity of the nanocomposites is measured according to the ASTM D5470 standard, using the setup shown in Fig. 4 (a), which has been designed and manufactured in-house. Samples with dimensions of 19 × 9 × 2 mm are cut from the molded samples and placed between hot and cold metal (copper) bars in the thermal conductivity measurement device. The setup is thermally isolated; it uses thermally insulating ceramic in order to have one-dimensional heat flow along the copper bars and the sample.

For measuring the thermal conductivity of the composites using the setup shown in Fig. 4 (a), a force is applied to the test fixture in the direction that is perpendicular to the test surfaces and maintains the parallelism and alignment of the surfaces while minimizing interfacial thermal resistances. The specimen temperature is maintained at 50 ± 1 °C by adjusting the heat flow, Q, generated from the electrical heater, and the temperature of the parts, the voltage, and the current applied to the heater are recorded during thermal conductivity measurements. The thermal impedance of the specimens is measured, and to measure the thermal impedance of a composite with different thicknesses, similar specimens are stacked to obtain the values versus thickness. In other words, first, the thermal impedance is measured for one layer, then for two

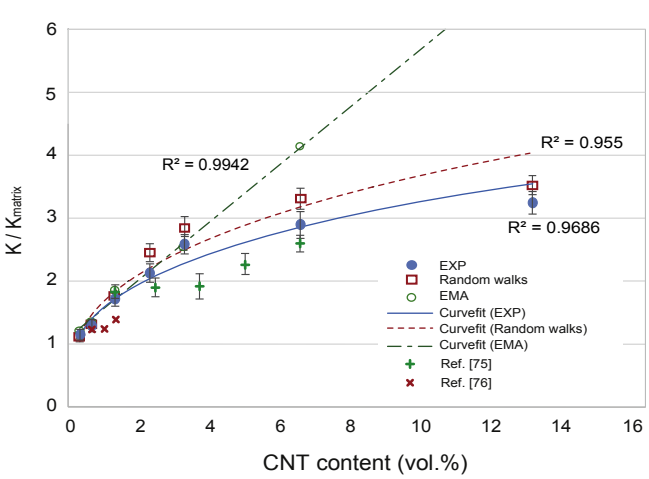

Fig. 5. Thermal conductivity of semi-aligned CNT nanocomposite specimens obtained from experiments and modeling (100 iterations).

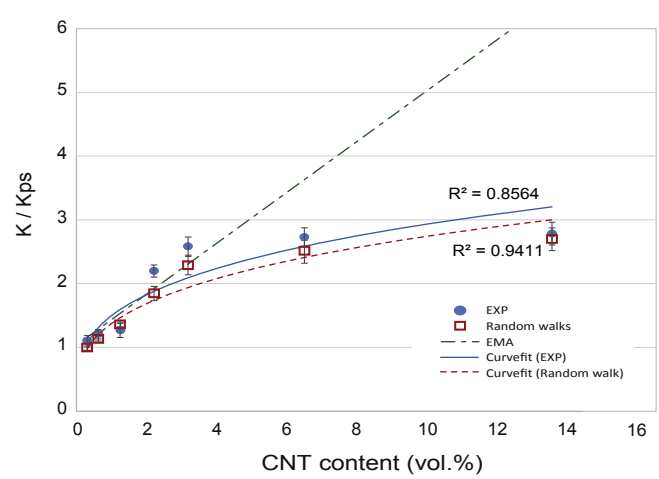

Fig. 6. Thermal conductivity of randomly-dispersed CNT nanocomposite specimens obtained from experiments and modeling (100 iterations).

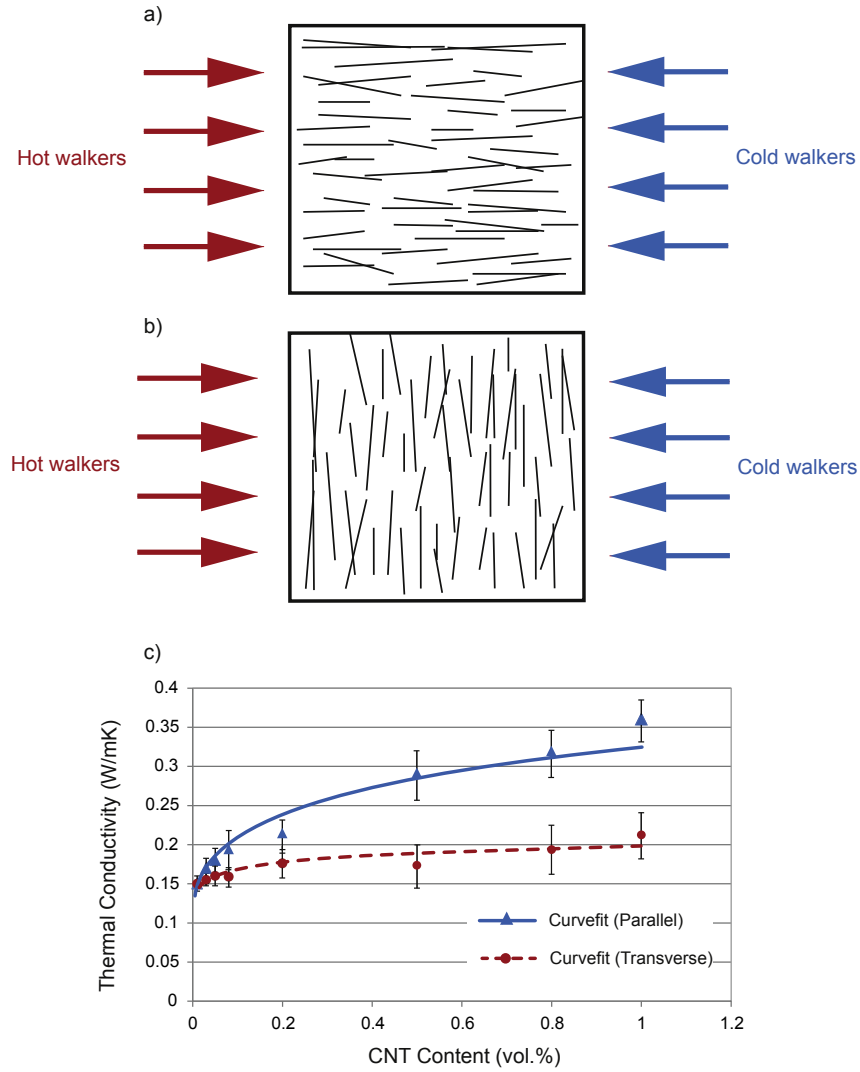

Fig. 7. Schematic view of two CNT-alignment directions: (a) parallel and (b) transverse to the travel direction of walkers and (c) comparison of thermal conductivities computed for both directions.

stacked layers, and finally for three layers stacked together. Fig. 4 (b) shows an example of the thermal impedance versus thickness of a single composite; data are fitted by a straight line whose slope is the reciprocal of the thermal conductivity of the composite [71–73]. For all the compositions, a linear regression method is used to depict the linear trend of the data. For each set of samples, an R-squared value is also calculated, and the average of the R-squared for the composites is 0.9888 (±0.0118).

The thermal impedance, $Z_{thermal}$ [K.m^2/W], is expressed by:

$$Z_{thermal} = \frac{A}{Q}(T_H - T_C) \tag{15}$$

where A is the area of the sample [m^2], Q is the heat flow [W], and T_C and T_H are the temperature of the cold and hot meter bar surfaces in contact with the specimen [K], respectively.

4. Results and discussion

4.1. Thermal conductivity model comparison of CNT/PS nanocomposites

Random walk simulations for thermal conductivity modeling are compared with those experiments considering the alignment of the CNTs within the polymeric matrix. Two different degrees of

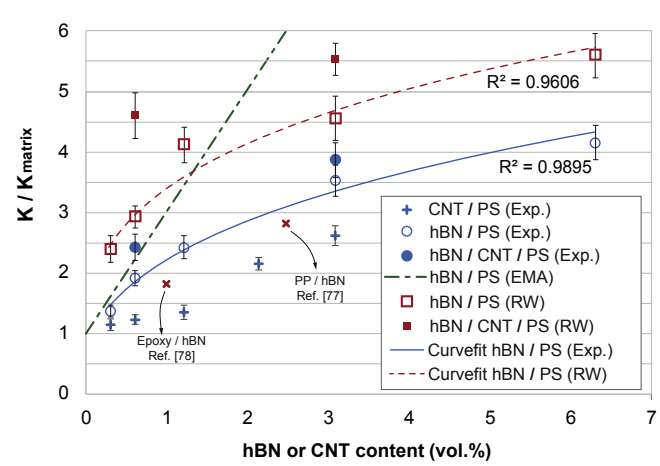

Fig. 8. Comparison of the thermal conductivity results obtained from EMA, random walk and experiments.

alignment (random dispersion and semi-aligned CNTs) are taken into account. The predicted thermal conductivity values are compared with the experimental results. It is well known that the thermal conductivity of the composites increases as the thermal boundary resistance at the CNT - matrix interface decreases. CNT distribution and the associated contact resistance also have a strong effect on the effective thermal conductivity, with higher conductivity associated with lower CNT - CNT contact resistance [74].

Figs. 5 and 6 compare results obtained from the random walk modeling, EMA, and experimental results. Experimental and random walk results are curve-fitted to the power law, and EMA results are fitted to linear lines. Fig. 5 shows the thermal conductivity values for specimens with semi-aligned CNTs compared to previously published works [75,76]. It is clearly shown that the relative thermal conductivity of composites increases with the addition of CNTs [76]. Thermal conductivities predicted from EMA are close to experimental measurements in low concentrations but way high in high concentrations. The EMA model represented in the paper does not reflect the effect of agglomeration of CNTs. In other words, the EMA model only considers geometry, thermal properties, and orientation of the entire CNTs. This assumption could be the reason for the increase in the thermal conductivity of composites with an increase in the concentration of CNTs. Meanwhile, the results from random walk modeling relatively agree well with the measured thermal conductivities in all concentrations. As it was described previously, injection-molding setup is used to fabricate specimens with semi-aligned orientation of CNTs. As shown in Table 4, injection molding is carried out at lower processing temperatures to produce semi-aligned composites. The deviation between random walk and experimental measurements may arise from the difference in the degree of alignment and distribution of CNTs. The orientation of CNTs is well-controlled in the model, whereas the alignment and distribution in the fabricated specimens cannot be controlled equivalently. Therefore, slightly higher thermal conductivity values are predicted by the random walk model, as CNTs are more oriented and able to transfer heat.

Fig. 6 compares predicted and measured thermal conductivities for random dispersed CNT composites. Similar to the semi-aligned composites, the predictions from random walk modeling approximate to the experimental measurements. Results of Fig. 6 show that the developed random walk model is more accurate for predicting thermal conductivity of randomly dispersed CNTs compared to that of semi-aligned ones. However, there are some mismatches at concentrations around 3 vol%, which requires further investigations through morphology of the nanocomposites to identify the cause.

The proposed computational model (Monte-Carlo simulations with the 3D random walk technique) is faster than molecular dynamics simulations and shows good agreement with the experimental data, as illustrated in Figs. 5 and 6. At low CNT volume fractions (approximately below 5 vol%), the thermal conductivity of the nanocomposite steadily increases with an increase in CNT content, irrespective of the distribution of CNT orientation. This is attributed to the fact that as the CNT density in the matrix increases, interconnected-CNT pathways are more available to the phonons (thermal walkers) to traverse through the nanocomposite. It is noted that the resistance due to contact between CNTs is negligible at such low CNT volume fractions and thus has no measurable impact on the nanocomposite thermal conductivity. However, at the volume fraction above approximately 4 vol%, the considerably increased amount of contact between CNTs resist the motion of phonons, while the increasing connectivity of the highly conductive CNT network propels the transfer of phonons across the nanocomposite. Due to these two competing influences, the thermal conductivity of the nanocomposite reaches a steady state, and the curve slopes change at around 4 vol% as shown in Figs. 5 and 6.

The influence of the CNT orientation distribution on the nanocomposite thermal conductivity is shown by comparing Figs. 5 and 6. At a given thermal boundary resistance and a CNT volume fraction, the random jump of phonons along the heat flow direction rises with an increase in number of the CNTs aligned with the flow direction, resulting in better heat transfer. Thus, the thermal conductivity should be highest when CNT nanocomposite is fully-aligned and lowest when CNT is randomly dispersed.

Interestingly, at high concentrations of CNT for randomly-dispersed nanocomposites (Fig. 6), experimentally measured values are greater than those predicted by the random walk model. It is speculated that the curved shapes of the CNTs and the created networks through their agglomeration in the fabricated specimens may have caused this mismatch, which is not taken into account in the model. Further investigation on the morphology of the fabricated samples is required in order to reveal it.

As it is noticed that alignment of CNTs may change the thermal conductivity of composites, effects of alignment of CNTs on thermal conductivity of composites are studied through random walk modeling. Creating a perfectly-aligned network of CNTs in fabricated composites could be very challenging; however, this can be performed easily through the developed random walk model. Three-dimensional RVEs with perfectly-aligned CNTs are attempted to calculate thermal conductivity in two alignment directions: parallel and transverse to the travel direction of hot/cold walkers as illustrated in Fig. 7 (a) and (b). CNTs are aligned in the RVE with azimuthal angle variation of smaller than 10. Fig. 7 (c) shows the results as a function of CNT contents. It is seen that thermal conductivity increases with CNT content for both alignment directions. The thermal conductivity increases more rapidly for the parallel direction than the transverse direction. At the CNT content of 1 vol %, the thermal conductivity of the parallel direction is higher than the transverse direction by a factor of 1.5. This is because more paths are provided to the thermal walkers when CNTs are aligned in the same direction to the travel direction of walkers.

4.2. Modeling of thermal conductivity of hBN/polymer nanocomposites

Thermal conductivity of the hBN/PS nanocomposites is calculated using both the random walk and EMA methods. These results are compared with the experimental results in Fig. 8 conducted in this research work and previously published ones [77,78]. Furthermore, to investigate the effects of the addition of CNTs to hBN/PS composites, two types of specimens with equal contents of hBN and CNT are fabricated. Thermal conductivity of these specimens are also calculated through the random walk model and compared with experimental results. For hBN/PS composites, results obtained from the EMA do not agree with the experimental values, especially in higher concentrations. This could be attributed to the fact that dispersion of the hBN nanoparticulates are considered as homogenized and random dispersed ($<Cos^2\theta> = 1/3$) in the EMA model. However, for the fabricated hBN nanocomposites, there could still be some agglomerations among hBN nanoparticulates. One can realize that predicted values using random walk modeling are higher than experimentally measured value, which contradicts with results of Fig. 6 with predicted values lower than experiments. This outcome could be attributed to the difference in the fillers geometry (1D CNTs versus 2D hBNs) and contact resistances (10^{-10} m^2 K/W for CNTs and 7×10^{-10} m^2 K/W for hBNs).

Hybrid-filled nanocomposites (with both hBN and CNT) have thermal conductivity higher than nanocomposites containing only hBN nanoparticulates. The high thermal conductivity of hybrid composites is achieved probably because of using both cylindrical and 2D shapes of nanoparticulates. When cylindrical inclusions like CNTs are combined with nanoplatelets such as graphene, CNTs can form an extended filler network by bridging neighbouring nanoplatelets [17,26,79–81]. Therefore, due to the fact that graphene and hBN fillers have similar shapes, the same effect could be expected.

In addition to the above, Fig. 8 includes results from random walk modeling. It shows that an increase of the hBN nanoparticulate content enhances thermal conductivity. However, higher thermal conductivity in all hBN concentrations are observed in the modeling, compared to experiments. The effect of hybrid compositions (combination of 1D CNTs and 2D hBNs) is reflected in the random walk modeling. Similar to the experimental results, hybrid composites of CNTs and hBNs have higher thermal conductivity values than those containing only hBN nanoparticulates. The disagreement between the random walk modeling and experiments could be attributed to better dispersion generated by a fully computational modeling within the RVE. However, as discussed earlier, hBN particulates are not completely separated during mixing nanocomposites, and there could be multilayers of nano-flakes instead of one layer of hBNs. Furthermore, agglomeration of hBNs by lower quality of mixing leads to a lower distribution of fillers [82].

The above facts can explain the influence of the distribution and content of hBNs on the thermal conductivity of the nanocomposites. Incorporating other effects, such as intermittent filler contact and agglomeration of the fillers in the matrix, in the random walk model could possibly predict the thermal conductivity of the nanocomposites more accurately.

5. Conclusions

In this paper, the thermal conductivity of CNT and hBN composites is investigated through theoretical and computational modeling and experimental measurements. A computational model using a random walk method is developed to predict thermal conductivity of CNT/PS composites with different alignments, concentrations, and aspect ratio. The same model is used for thermal conductivity of hBN/PS composites with a random dispersion and aspect ratio of ellipsoidal inclusions. To evaluate the developed model, thermal conductivities are measured from fabricated nanocomposites. An injection molding technique and compression molding are utilized to fabricate nanocomposites with semi-aligned and randomly dispersed CNTs to investigate the effect of CNT alignment. It is observed that alignment of CNTs may result in higher thermal conductivity values. The developed random walk model is more precise in predicting thermal conductivity values of composites with random dispersion of CNTs. The estimations from the EMA model are higher than experimental measurements in the high CNT concentrations; however, they are in good agreement in the low concentrations. This could be attributed to a limitation in EMA modeling, which does not consider agglomeration of CNTs. Hybrid compositions of CNT and hBN are also studied in this paper. Both experimentally measured and stochastically predicted results suggest that hybrid composites may have higher thermal conductivities. This outcome could be attributed to bridging between hBN nanoparticulates (two dimensional) by CNTs (one dimensional).

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